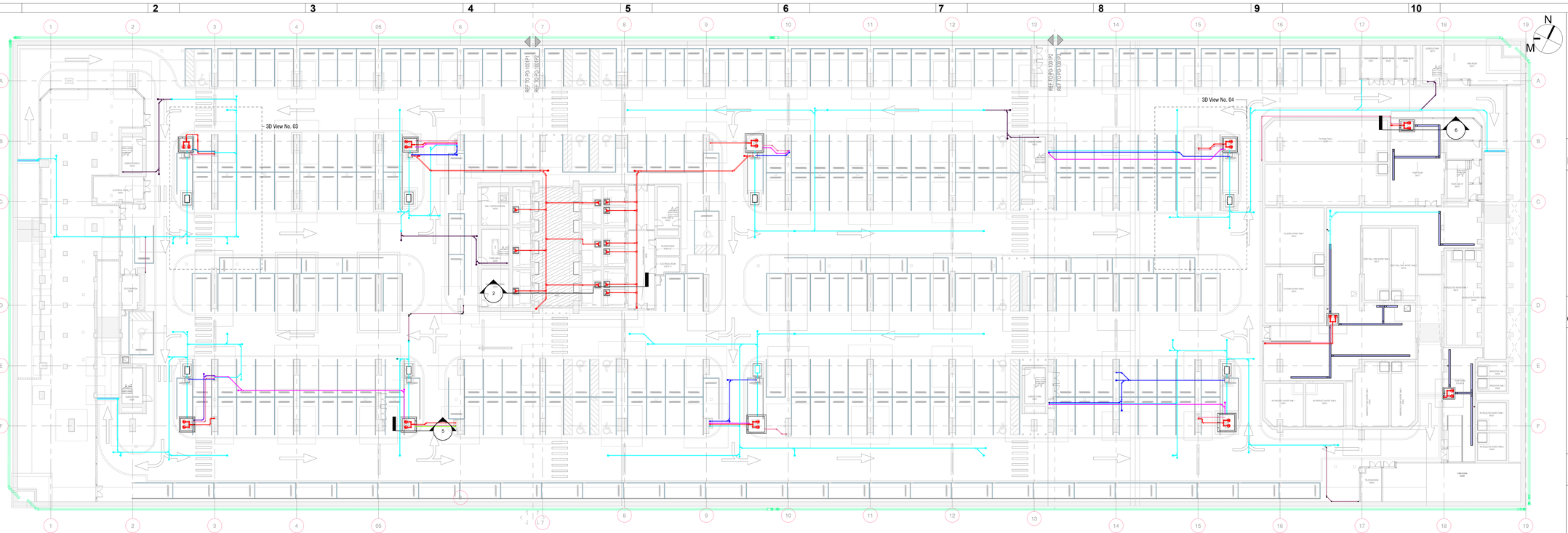
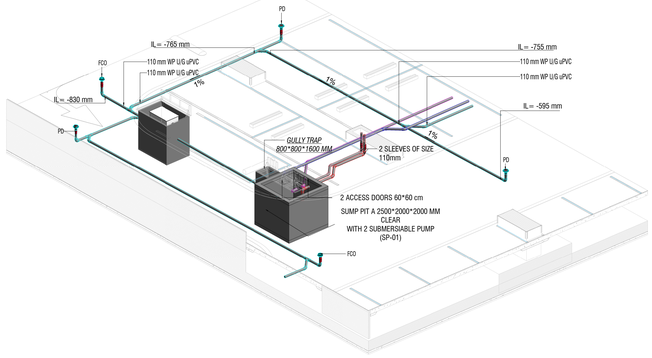
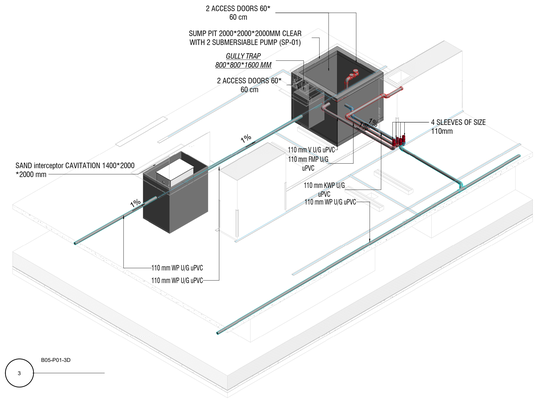


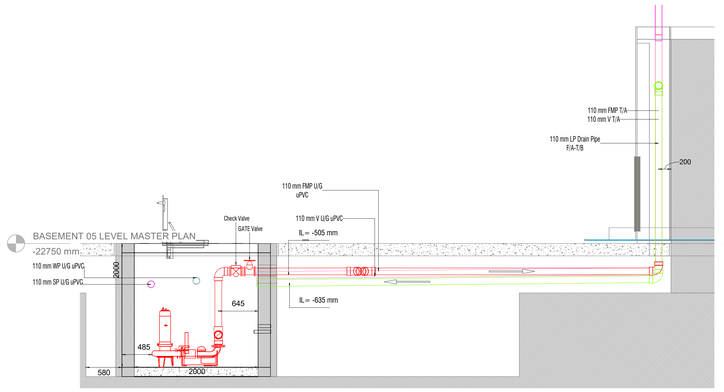
1 BASEMENT 05 LOWER LEVEL MASTER PLAN
1 : 250



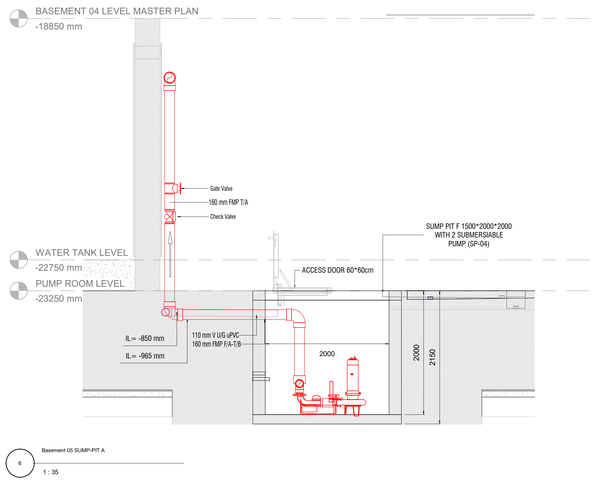
4 B05-P03-3D



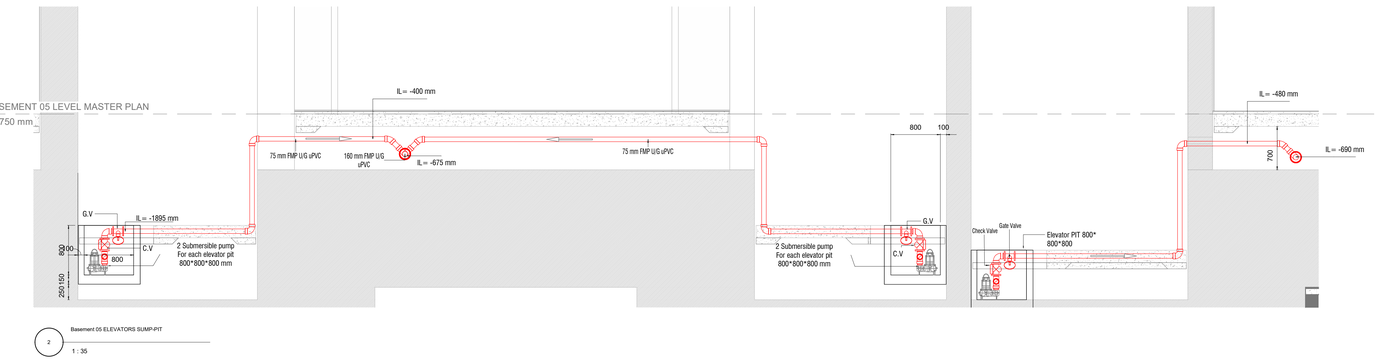
3 B05-P01-3D



5 Basement 05 SUMP-PIT F
1 : 35



6 Basement 05 SUMP-PIT A
1 : 35



2 Basement 05 ELEVATORS SUMP-PIT
1 : 35

CLIENT

PROJECT CONSULTANT

KEY PLAN



GENERAL NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE NOTED.
2. EXACT LOCATIONS OF ALL FLOOR DRAINS AND LINEAR DRAINS TO BE COORDINATED WITH ID DRAWINGS.
3. FOR ALL HORIZONTAL DRAINAGE PIPES: A MINIMUM SLOPE OF 2% SHOULD BE PROVIDED FOR PIPES 100 mm DIAMETER AND SMALLER, AND 1% FOR PIPES 100 mm DIAMETER AND LARGER.
4. DESIGN AND INSTALLATION OF PLUMBING SYSTEMS SHALL BE IN COMPLIANCE WITH SBC 701.
5. THE INDICATED DIAMETER IN THE DRAWING IS THE NOMINAL DIAMETER.
6. ALL THE SLEEVES SIZES AND DIMENSIONS WILL BE INDICATED IN THE BUILDER MARKS SHEET.
7. REGARDING DRAIN POINTS:
   -THE MINIMUM APPLICABLE SLOPE SHALL BE PROVIDED ON SITE TO IMPROVE WATER RUNOFF AND MINIMIZE SAG POINTS AND AREAS OF WATER ACCUMULATION.
8. THE INDICATED DRAINAGE IN THE SHEET REPRESENT THE HIGHER LEVEL DRAIN OF LEVEL BELOW AND THE DRAIN FIXTURES OF THE CURRENT LEVEL

LEGEND & ABBREVIATION

SYMBOL	ABBREVIATION
U/G	UNDER GROUND
H/L	HIGH LEVEL
F/A T/B	FROM ABOVE TO BELOW
T/A	TO ABOVE
FFL	FINISH FLOOR LEVEL
IL	INVERT LEVEL
FD	FLOOR DRAIN
FCO	FLOOR CLEAN OUT
CO	CLEAN OUT
TD	TRENCH DRAIN
PD	PARKING DRAIN
WP	WASTE PIPE
SS	SOIL STACK
VP	VENT PIPE
TWP	TOILET WASTE PIPE
C.V	CHECK VALVE
G.V	GATE VALVE
PP	PERFORATED PIPE
BFS	BEFORE FINISH SLAP
FMP	Forced Main Pipe
▬	Trench Drain
■	Floor Drain - Rectangular
⊙	Floor Drain - Multi Inlets
⊙	Roof Drain
⊙	Park Drain
○	Floor Clean out
•	Clean Out
	Soil Pipe
	Toilet Waste Pipe
	Drain Waste Pipe
	SP DP Waste Pipe
	LP DP Waste Pipe
	Kitchen Waste Pipe
	Forced Drain Pipe
	Vent Pipe
	Condensate Pipe
	Perforated Pipe

DRAINAGE SYSTEMS AND THEIR PIPE MATERIAL

DRAINAGE SYSTEM	PIPE MATERIAL
INTERNAL DRAINAGE & VENT ABOVE GROUND (GRAVITY)	UPVC
CONDENSATED DRAINAGE (GRAVITY)	HDPE
UNDER GROUND DRAINAGE AT B05	HDPE PRESSURE PIPE
INTERNAL DRAINAGE (PRESSURIZED)	UPVC
STORM DRAINAGE	UPVC, SCH40
LABORATORIES DRAINAGE	POLYPROPYLENE
KITCHEN DRAINAGE	HDPE

ARTAR COORDINATION

CHECKED	NAME	SIGNATURE	DATE
PRINCIPAL MECHANICAL ENGINEER	Eng.Yaser Al Hafiz		
PRINCIPAL MECHANICAL ENGINEER	Eng.Faras Elouah		
ELECTRO-MECHANICAL MANAGER	Eng.Mamoun Jawad		

MAAP COORDINATION

CHECKED	NAME	SIGNATURE	DATE
OPERATIONAL MANAGER	Eng.Ola Magdy		
BIM MANAGER	Eng.Mohammed Hany		
ARCHITECURAL & ID	Eng.Mohammed Hany		
STRUCTURE	Eng.Mahmoud Mounir		
ELECTRICAL & LC	Eng.Aly AlKassi		
MECHANICAL /FF/PL	Eng.Mohammed Kamal		

RELATED DOCUMENTS, LETTERS & RFI'S REFERENCES

NO.	DOC. NO.	DOC. DESCRIPTION	REV	DATE
01	DB142-DAR-TD-B01-DWG-PD-1001M	BASEMENT 05 LEVEL DRAINAGE MASTER PLAN	00	OCT - 2024
02	DB142-DAR-TD-B01-DWG-PD-1001P1	BASEMENT 05 LEVEL DRAINAGE PART 1	00	OCT - 2024
03	DB142-DAR-TD-B01-DWG-PD-1001P2	BASEMENT 05 LEVEL DRAINAGE PART 2	00	OCT - 2024
04	DB142-DAR-TD-B01-DWG-PD-1001P3	BASEMENT 05 LEVEL DRAINAGE PART 3	00	OCT - 2024
05	SMC 3 - RFI - PL 001	Drainage Study For Sumpits Between Axe 10&16	00	JUL - 2026
06	SMC 3 - RFI - PL 002	Access Doors & Wall Openings For Water Tanks	00	MAR - 2026
07	SMC 3 - RFI - PL 004	Holes Locations	00	APR - 2026
08	SMC 3 - RFI - PL 006	Flushing And Water Tanks	00	APR - 2026
09	SMC 3 - RFI - PL 008	Elevator Pits Are Connected To Soil Pump Pit	00	JUL - 2026
10	SMC 3 - RFI - PL 009	Basement Drain Points	00	JUL - 2026
11	SMC 3 - RFI - PL-AR-001	Pump Room And Tanks Level	00	JUL - 2026

REVISION

REV	DATE	ISSUED FOR	DRW	CHK	PM	BIMM
00	10.2026	SHOPDRAWING	A.H/O.S	A.K.	M.K.	T

PROJECT TITLE

SPECIALIZED MEDICAL CENTER – 3 (SMC 3)

RIYADH – KSA – AL WADI DISTRICT

B01 - MAIN HOSPITAL

DISCIPLINE
Drainage System

SHEET TYPE
Drainage System Plans - BIM DRAWING - LOD-350

SHEET TITLE
BASEMENT 05 USOG LEVEL DRAINAGE MASTER PLAN

PROJECT NO	BLDG. NO	SHEET NO	SCALE	REV MARK
SMC3	B01	1001	As indicated	00

DRAWING NO
SMC3-ARTAR-M-B5-PD-1001-DWG

0 10mm 50mm 100mm
(Sheet Size - ISO - A0 (841x1189 mm))